Single-shot measurement of reverse saturable absorption with self-seeded beam

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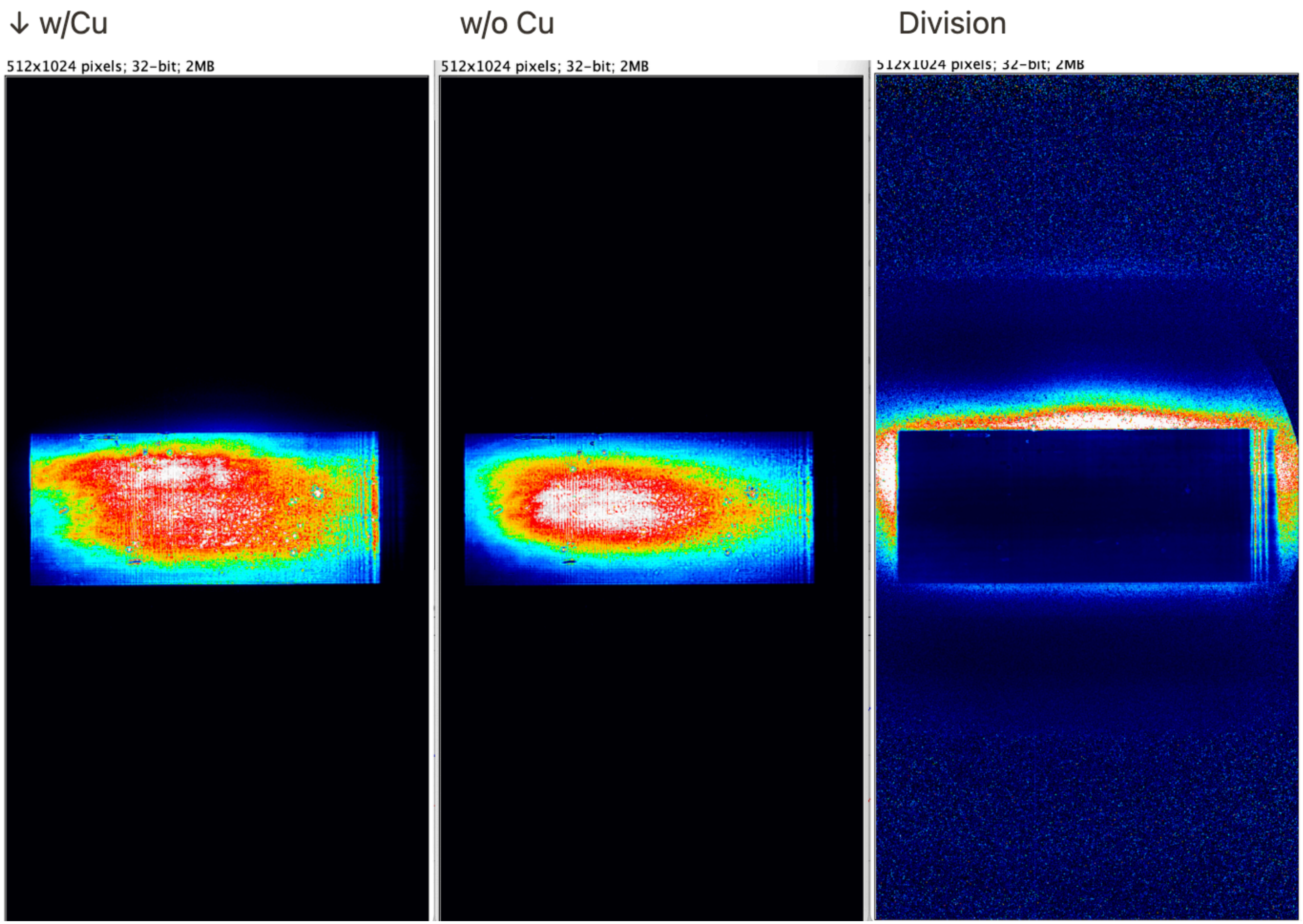
Reverse saturable absorption of copper induced by SASE pulses

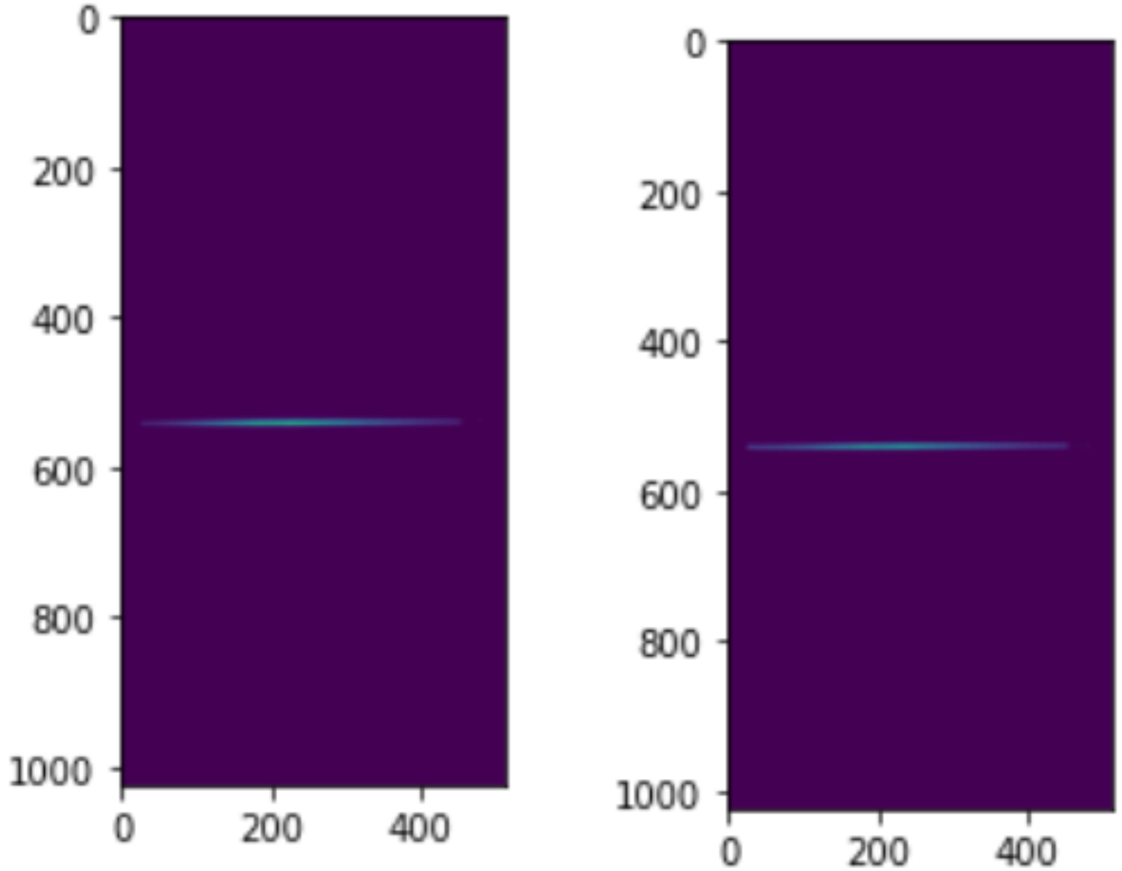


Si 220 flat crystal spectrometer

Beam divergence after transmitting copper foil seems different from original value

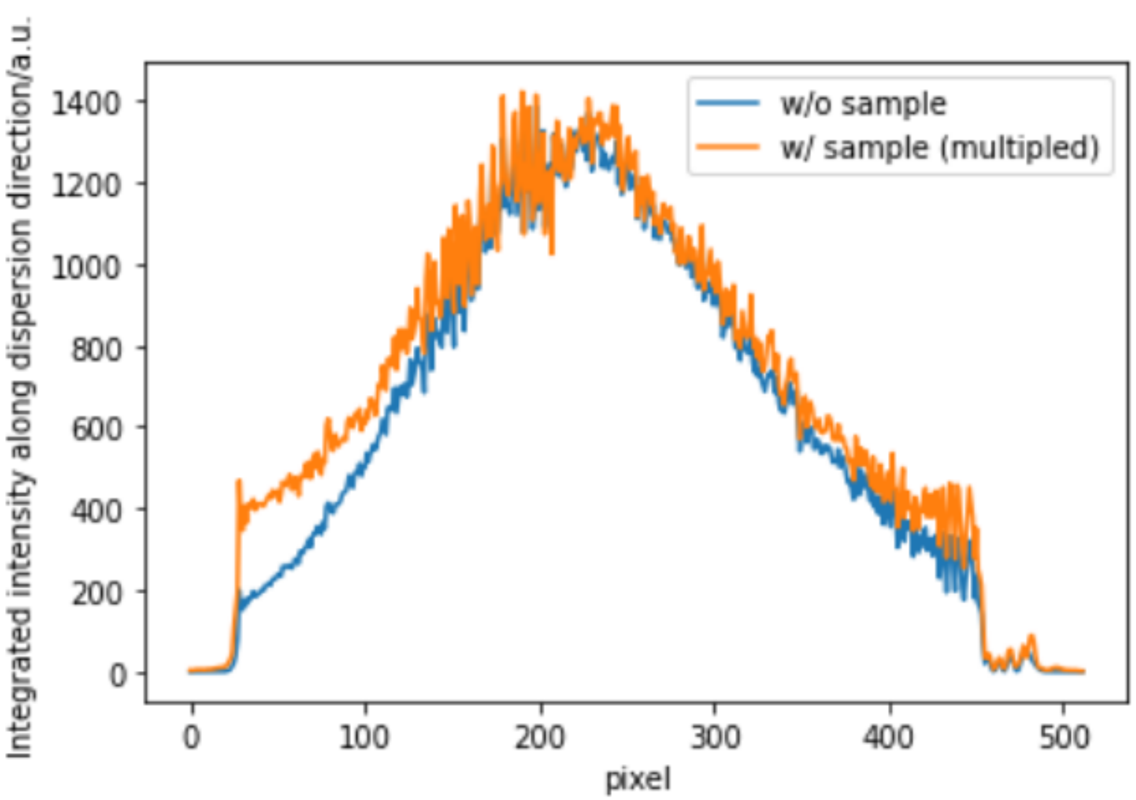
-> Crystal spectrometer might not be a good choice for precise evaluation of the transmittance.

RSA of copper induced by self-seeded pulses



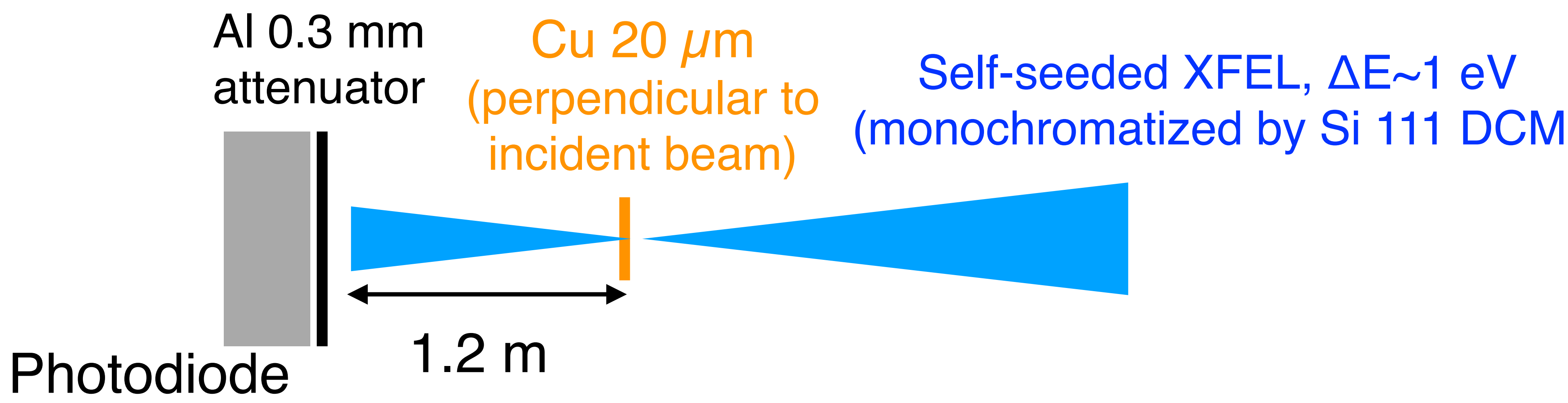
8.048 keV=Ka1

Si 220 flat crystal spectrometer



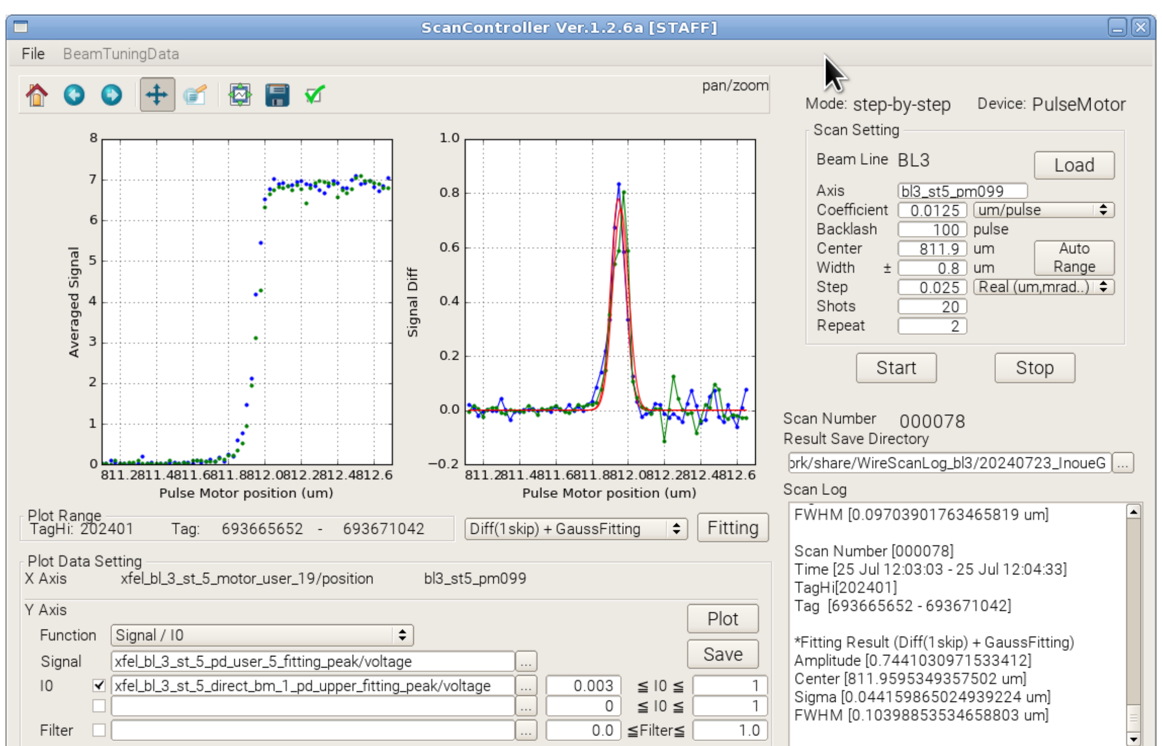
Beam divergence becomes large after transmitting copper foil

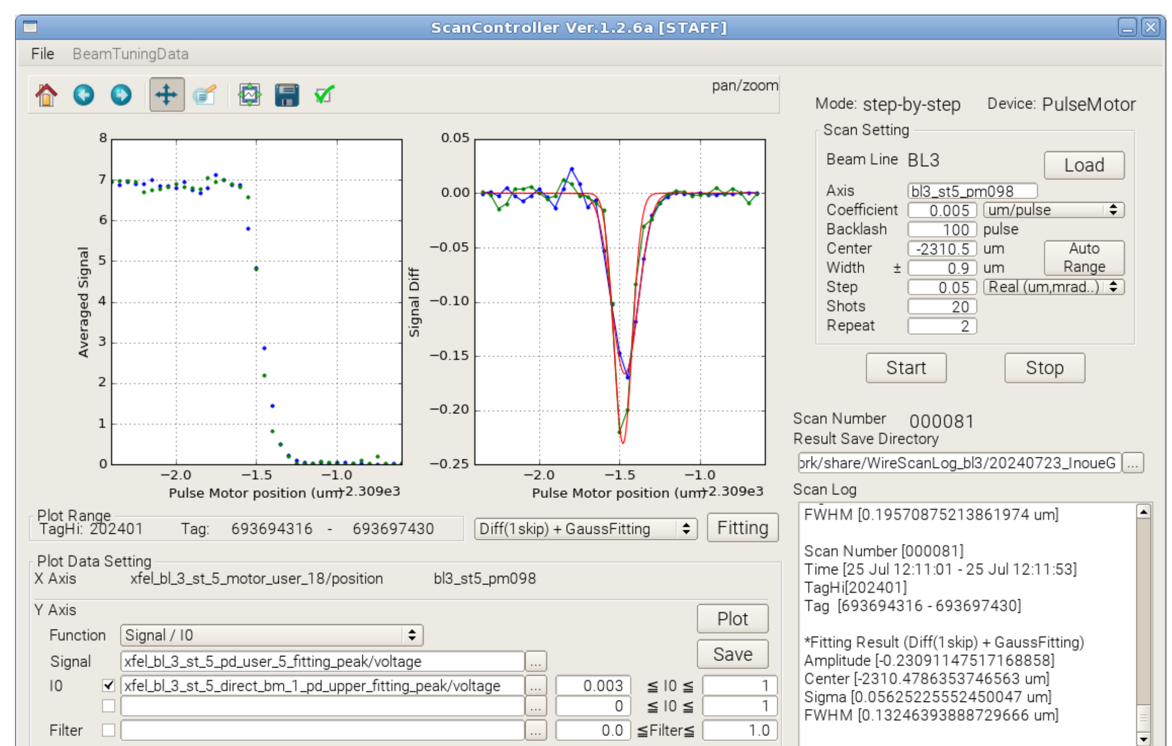
Setup



cf.
Cu foil was placed in front of photodiode
when measuring transmittance of cold state.

Beam size (FWHM): 100 nm (H) x 170 nm (V)

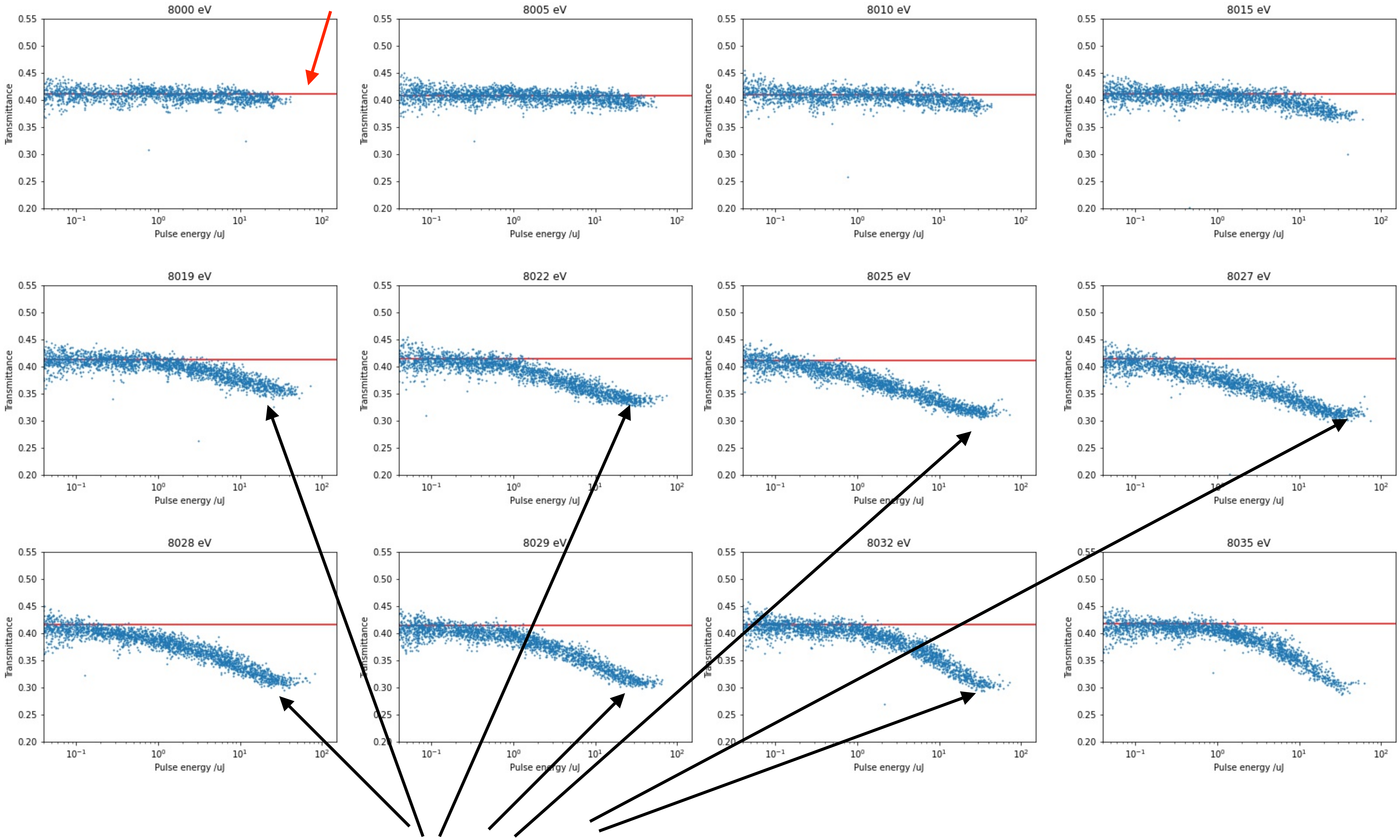




Results for Cu 20 um foil (photon energies around Ka2=8028 eV)

transmittance
for cold state (experimentally measured)

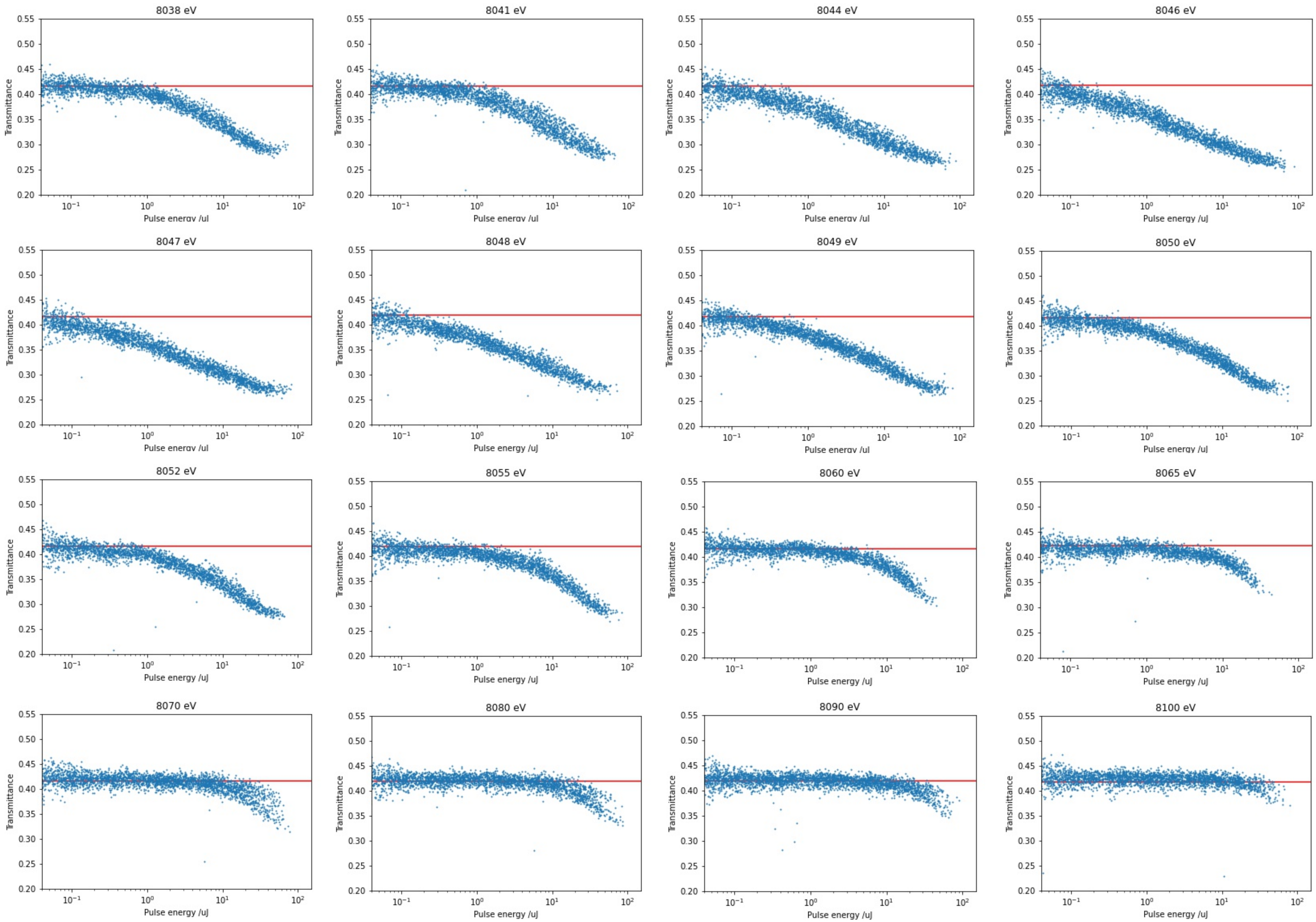
Each dot corresponds to single pulse



Reverse saturable absorption reached “saturation” (transmittance ~0.32@8028 eV)

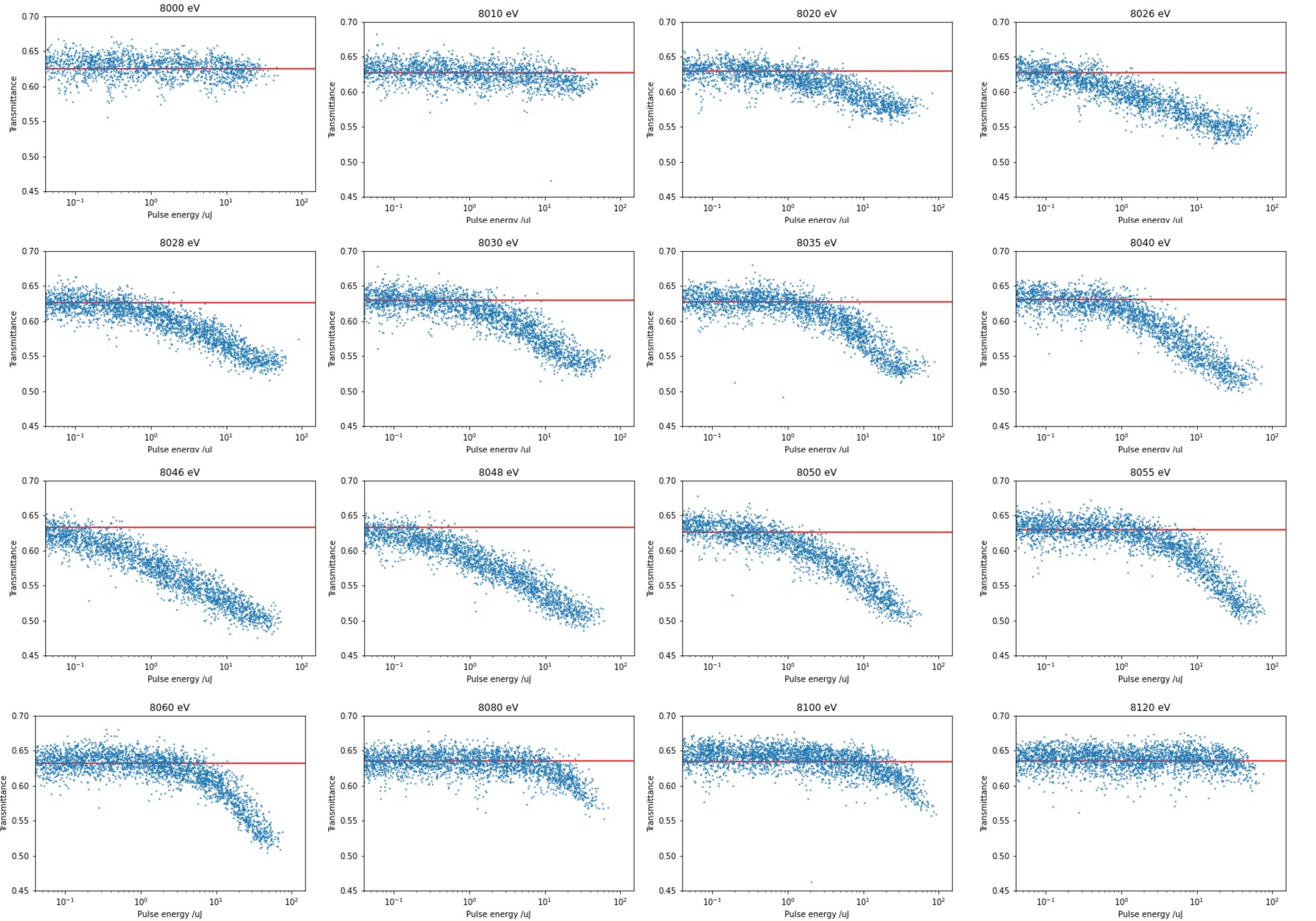
RSA was observed for photon energies higher than 8010 eV.

Results for Cu 20 um foil (photon energies around Ka1=8048 eV)



“saturation” value of transmittance was ~0.28@~8048 eV
RSA was observed for photon energies lower than 8090 eV.

Results for Cu 10 um foil:



Change in imaginary part of atomic scattering factor

Excess linear absorption coefficient due to reverse saturable absorption@8.048 keV:

$$u_{RSA} = -\log(0.5/0.65)/10\ \mu\mathrm{m}$$
$$= 260\ \mathrm{cm}^{-1}$$

Excess absorption cross section per atom:

$$\sigma_{atom} = \frac{M}{\rho N_A} u_{RSA} = 3.2 \times 10^{-21} \mathrm{cm}^2$$

M: molecular mass (63.55)
ρ: density (8.96 g/cm^3)
N_A: Avogadro constant (6.02 x 10^{23})

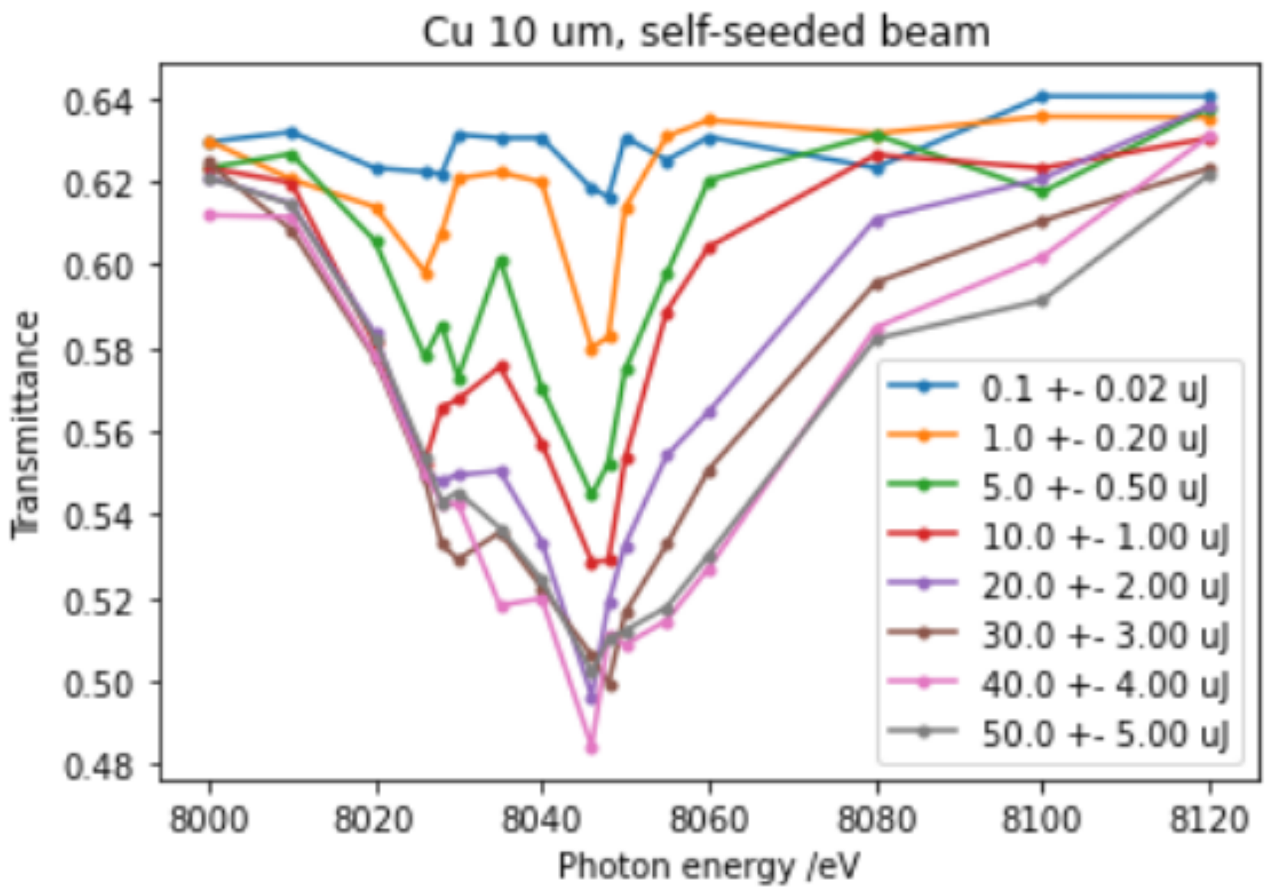
According to optical theorem, change in the imaginary part of atomic scattering factor may be given by

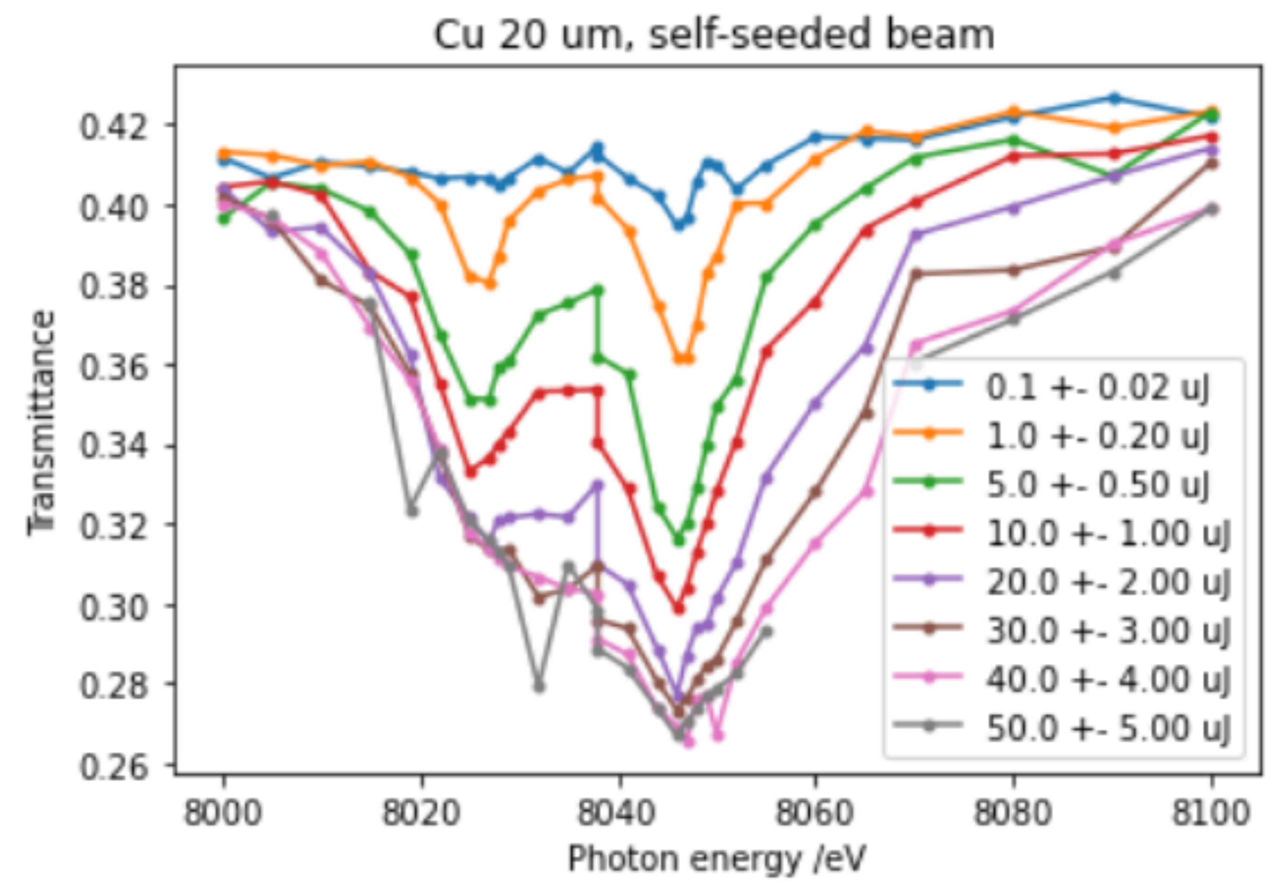
$$f'' = \frac{-k}{4\pi r_0} \sigma_{atom} = -0.37$$

k: wave number
r_0: classical electron radius

Enhancement of atomic scattering factor by transient resonance will be very small (but might be useful for phasing)

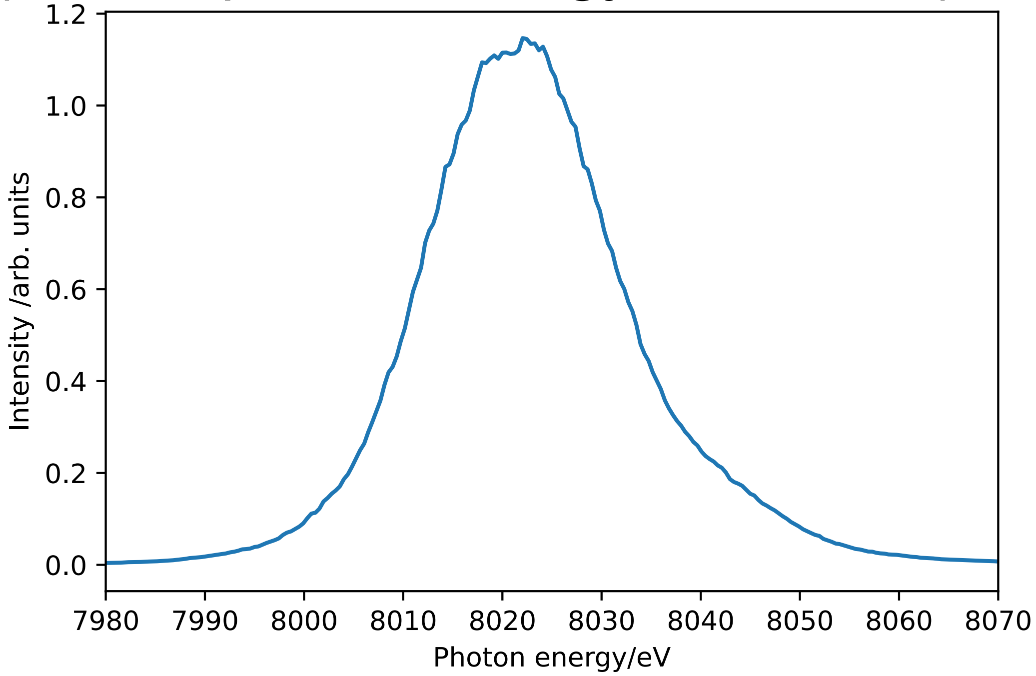
Transmittance of Cu for monochromatized seeded beam (ΔE~1 eV)

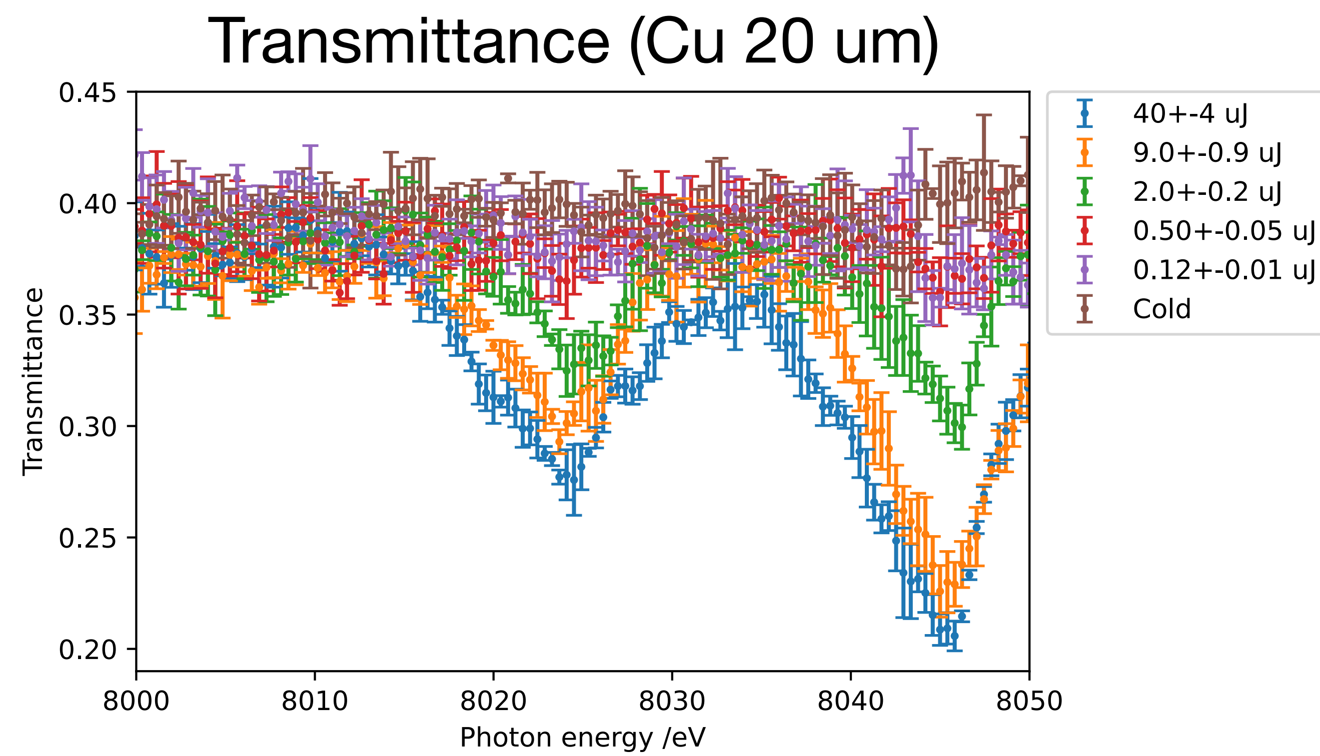




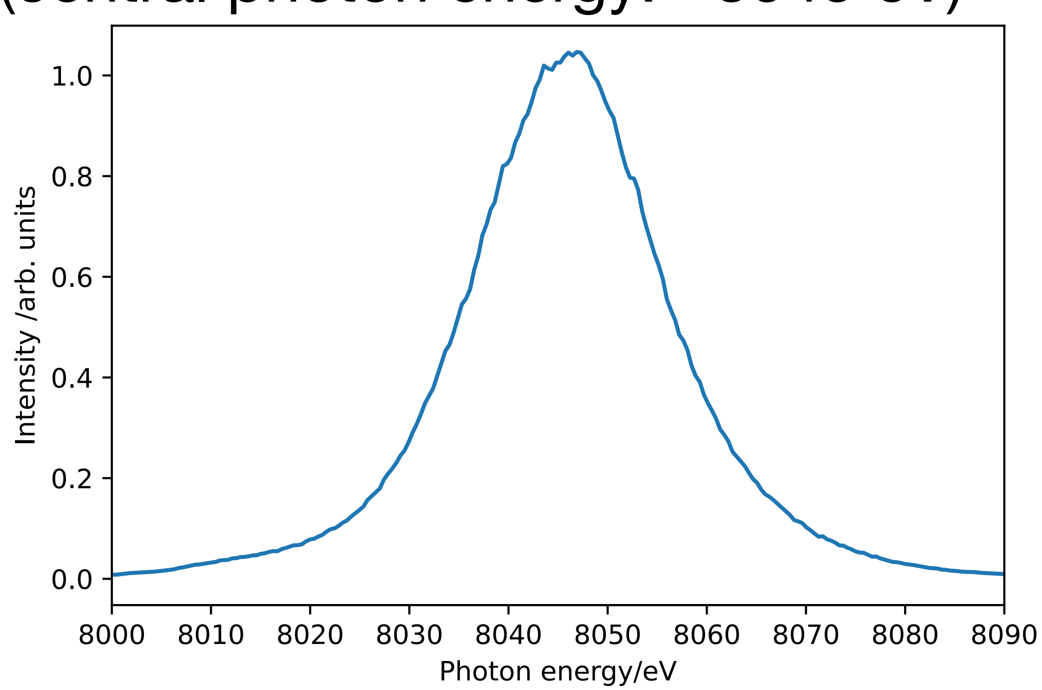
cf. reverse saturable absorption measured with SASE beam

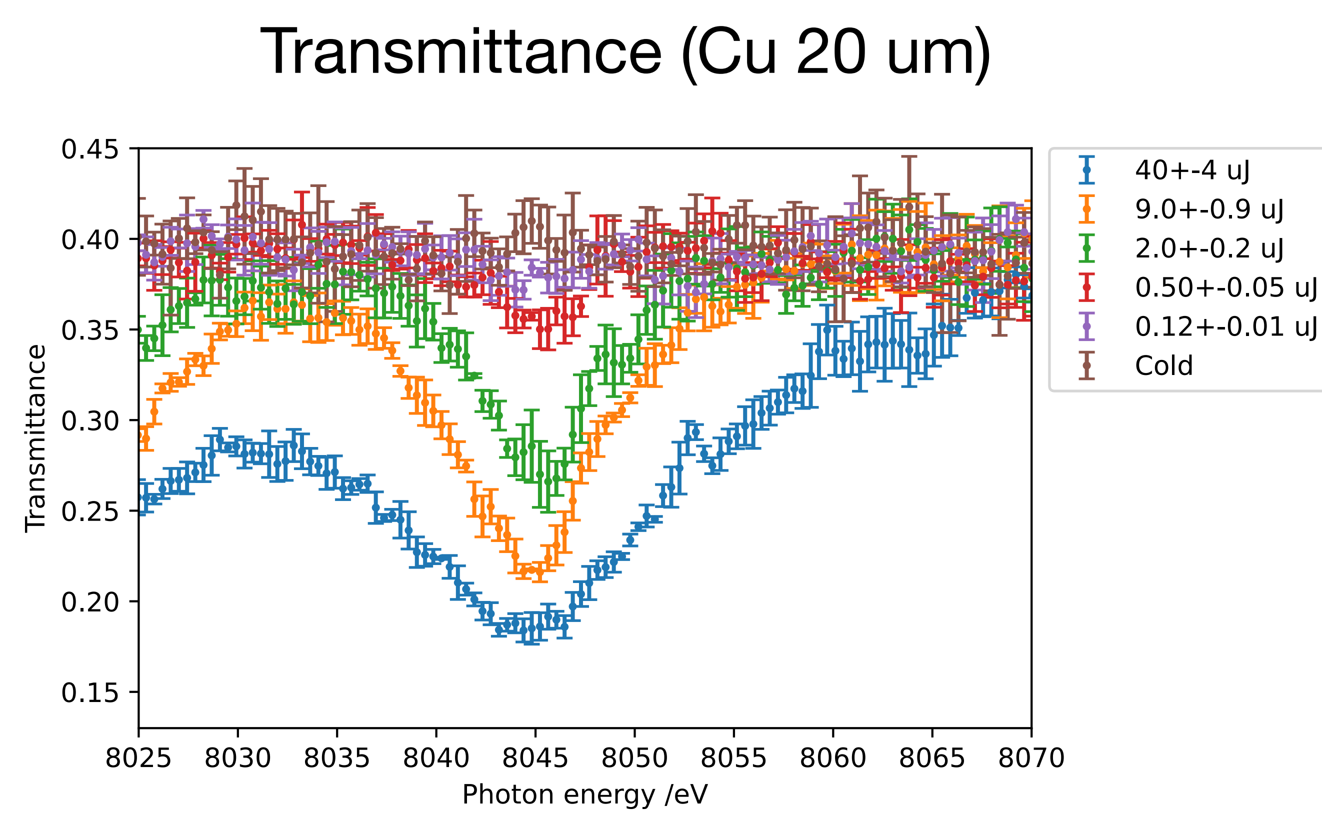
Average spectrum of incident beam
(central photon energy: ~8020 eV)



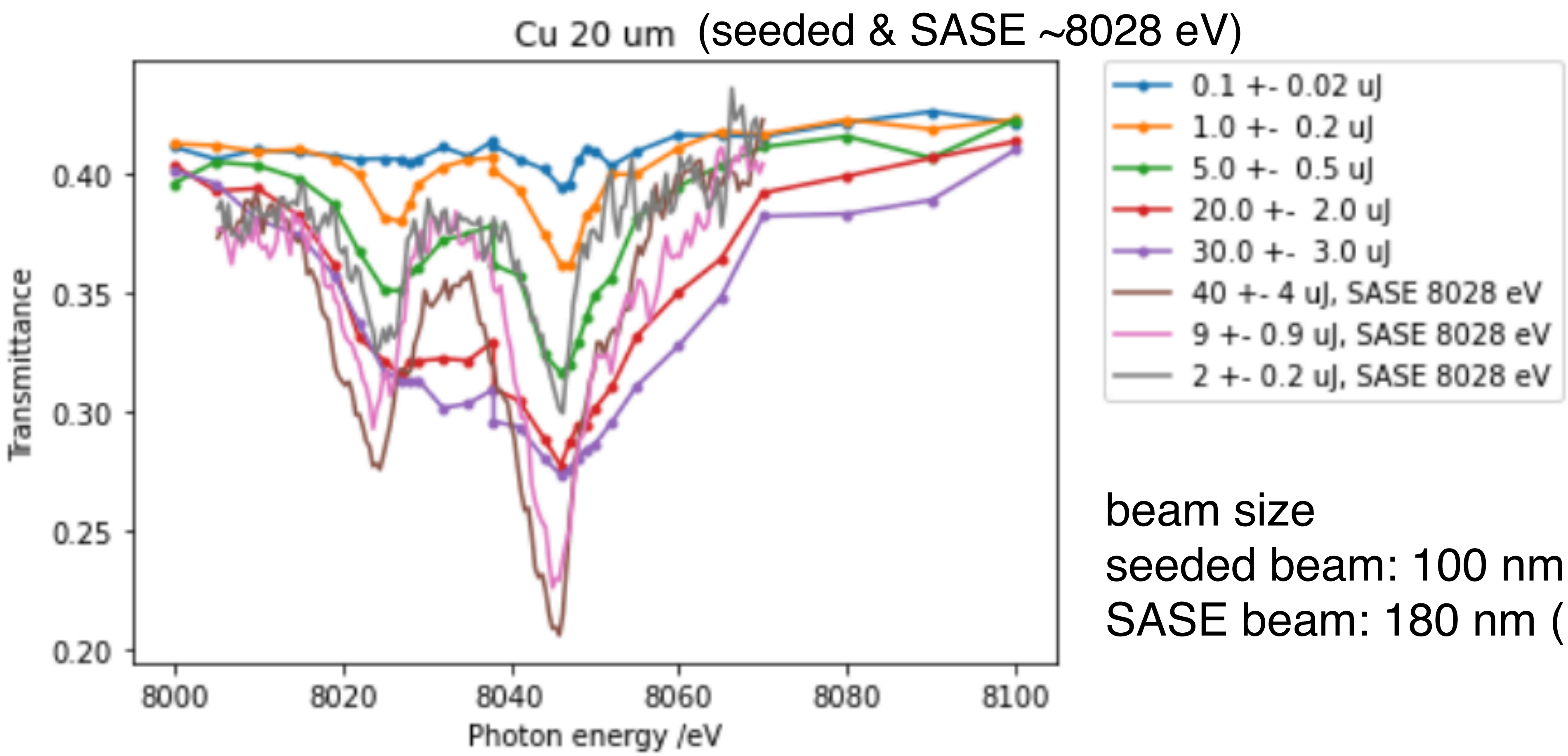


Average spectrum of incident beam
(central photon energy: ~8045 eV)

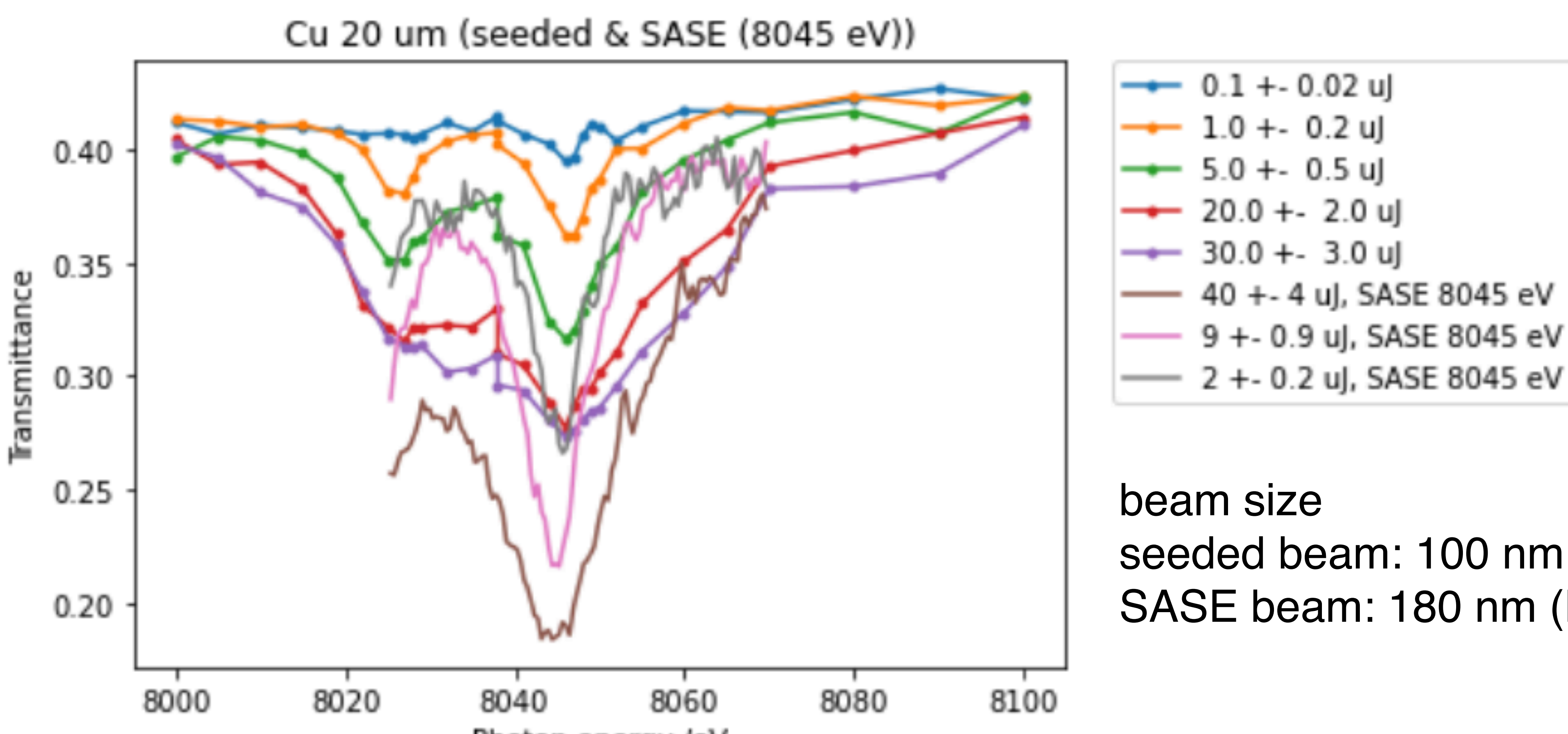




Data for seeded and SASE beams in one plot



beam size
seeded beam: 100 nm (H) x 170 nm (V)
SASE beam: 180 nm (H) x 220 nm (V)



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